

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			Its structure contains 32 faces Has a relative molecular mass of 60 Is an allotrope of carbon Has a giant ionic structure is a hydrocarbon compound It is a smart material Is 1 x 10²⁵ time smaller than a football ☒ ☐ ☒ ☐ ☐ ☐ ☒ award (2) for all 3 correct award (1) for any 2 correct award (1) for 3 correct and 1 incorrect if four boxes ticked award (0) for 2 correct and 2 incorrect if four boxes ticked	2			2		
	(b)	(i)		90 (2) award (1) for 180 or for the total number of sides on all pentagons or all hexagons i.e. (12 × 5) / 60 / (20 × 6) / 120 no ECF if incorrect number of sides calculated in first step	1	1		2	2	
		(ii)		6.97 × 10⁻²⁸ (3) award (2) for correct calculation of 0.697 / 0.69655 if incorrect answer award (1) for correct identification of radius as 0.55 if final calculation is totally incorrect award (2) for correct answers using the diameter instead of radius (5.57 × 10⁻²⁷) award (1) for answer not given in standard form (5.57) accept correct answers using π value from the calculator	1	2		3	3	